

# WHAT IS DIGITAL IMAGE PROCESSING

- Digital Image Processing means processing digital image by means of a digital computer.

## PURPOSE OF DIGITAL IMAGE PROCESSING

divided into 5 groups.

- Visualization – Observe the objects that are not visible
- Image sharpening and restoration – To create a better image
- Image retrieval – Seek for the image of interest
- Measurement of pattern – Measures different objects in an image
- Image Recognition – Separate, the objects in an image.

## DEFINE DIGITAL IMAGE

- A **digital image** is an **image** composed of **picture** elements, also known as pixels, each with finite, discrete quantities of numeric representation for its intensity or gray level that is an output from its two-dimensional functions fed as input by its spatial coordinates denoted with  $x, y$  on the  $x$ -axis and  $y$ -axis

## ELEMENTS OF VISUAL PERCEPTION

### • PERCEPTION

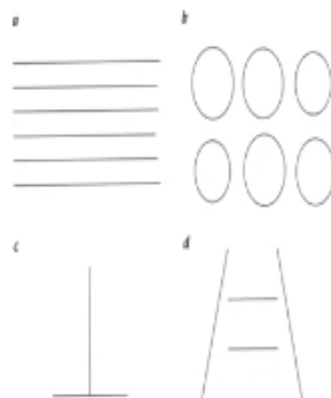
\*We can't think of image processing without considering the human vision system.

\*We observe and evaluate the images that we process with our visual system

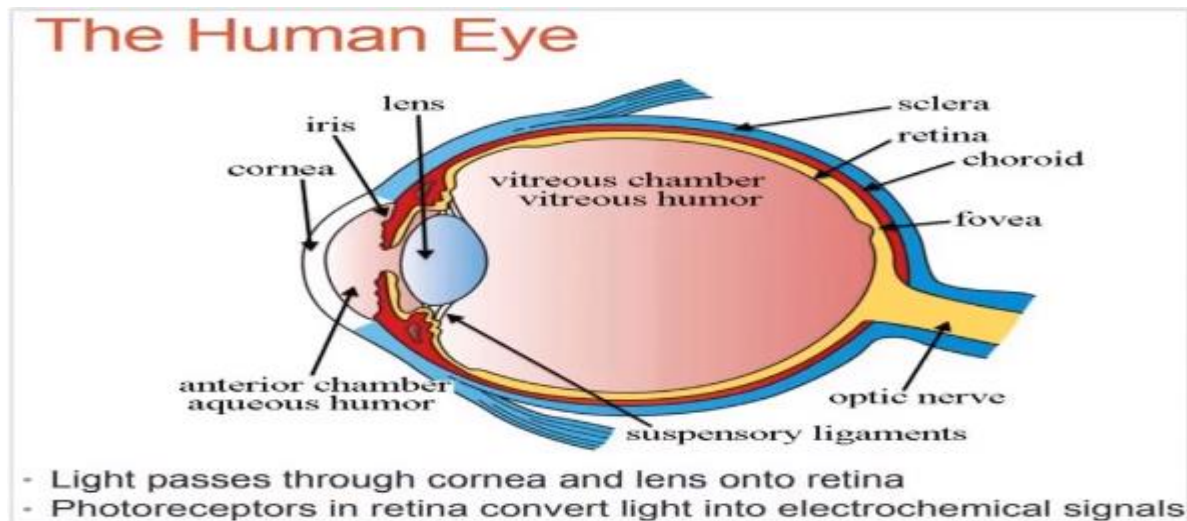
### Test images

• Test images for distances and area estimation:

- a) Parallel lines with up to 5% difference in length.
- b) Circles with up to 10% difference in radius.
- c) The vertical line appears longer but actually has the same length as the horizontal line.
- a) Deception by perspective: the upper line appears longer than the lower one but actually have the



# STRUCTURE OF HUMAN EYE

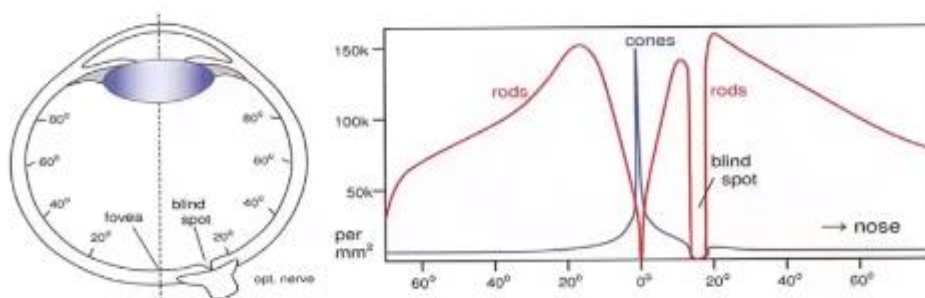


## Cross section of the eye

\*Blind spot B the absence of receptors area.

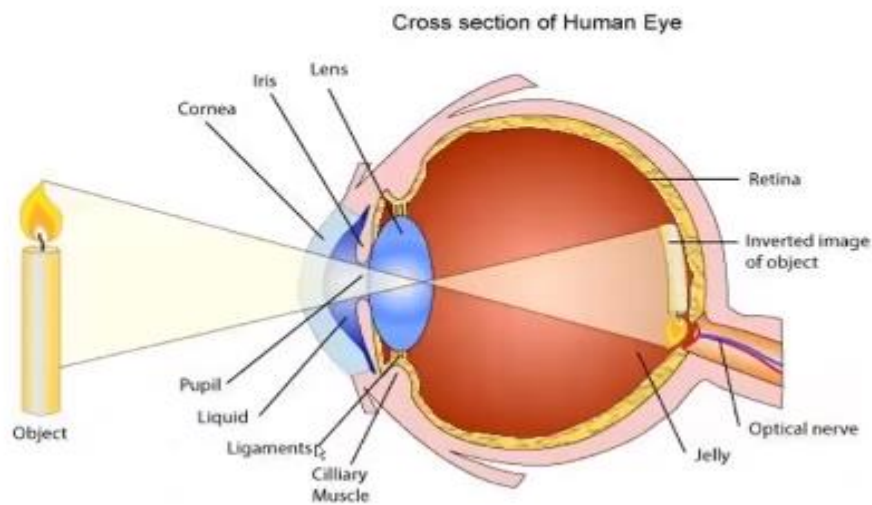
\*Cones are most dense in the center of the retina (in the area of the fovea).

## Distribution of Rods and Cones



- Highest density of cones around fovea
- Blind spot at Optic Nerve exit point

# Structure of the Eye



## Image Formation in the Eye

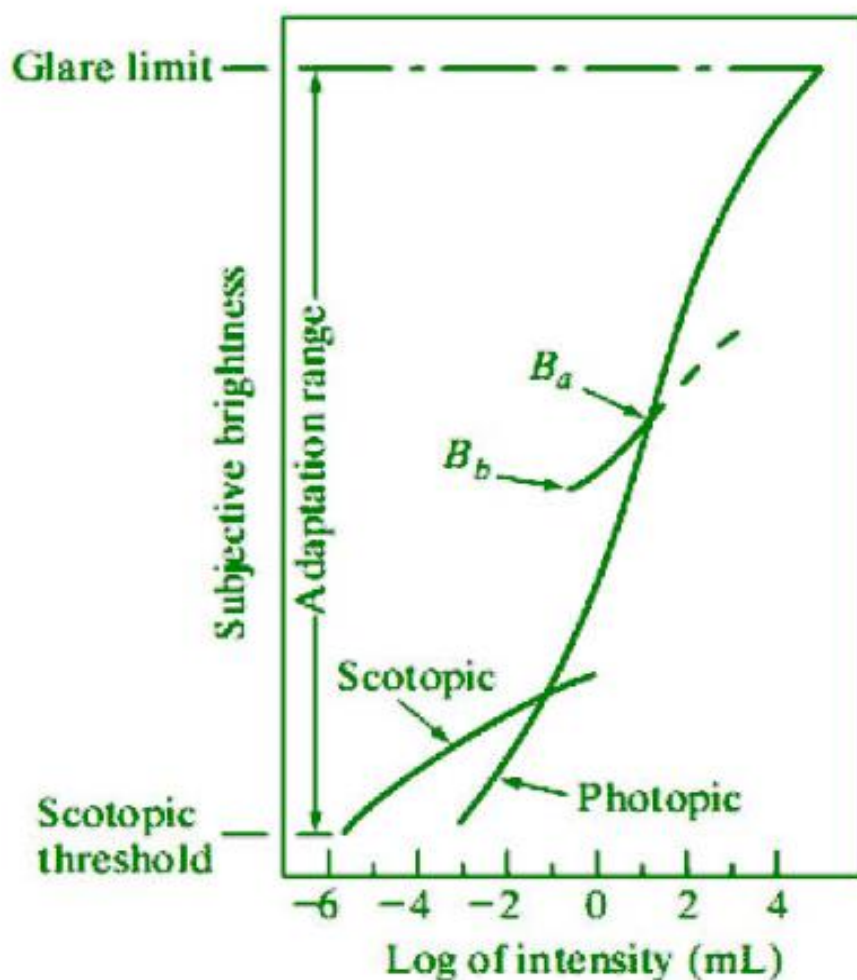
- Focal Length:
  - is the distance between the center of the lens and the retina
  - Varies from 17mm to 14mm ( refractive power of lens goes from minimum to maximum).
  - Object farther than 3m use minimum refractive lens powers(vice versa).

## Brightness Adaptation and Discrimination

- The ability of the eye to discriminate between brightness levels is an important consideration in creating and presenting images.
- The range of light intensity levels to which the human visual system can adapt is on the order of 10<sup>10</sup> from the scotopic threshold(minimum low light condition) to the glare limit(maximum light level condition)

# Brightness Adaptation of Human Eye

- **Brightness adaptation** is a phenomenon which describes the ability of the human eye in simultaneously discriminating distinct intensity levels.
- **Brightness adaptation level** is the current sensitivity level of the visual system for any given set of conditions.
- The **simultaneous contrast** is a phenomenon which describes that the perceived brightness of a region in an image is not a simple function of its intensity rather it depends on the intensities of neighboring regions.
- **Subjective brightness** is brightness as perceived by the human visual system and is a logarithmic function of the light intensity incident on the eye.



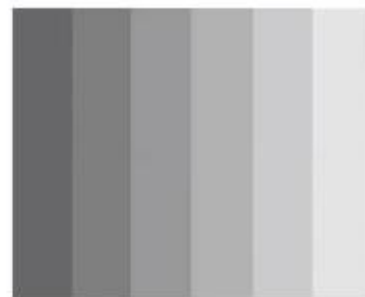
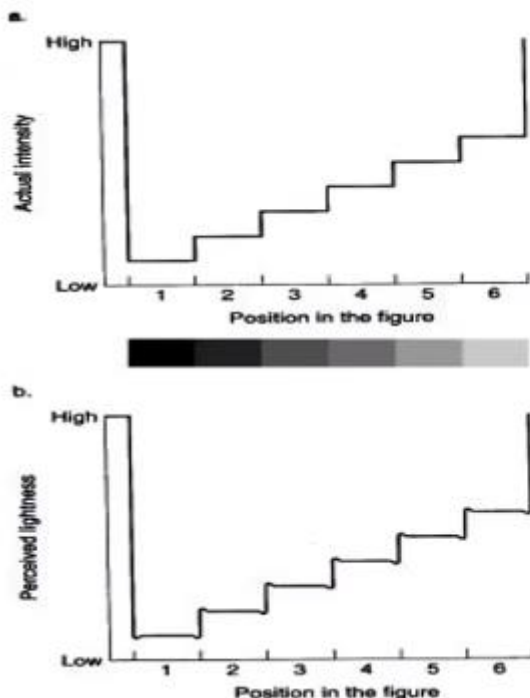


# Brightness adaptation

- For a given set of conditions, the current sensitivity level of the visual system is called the brightness-adaptation level which may correspond, for example, to brightness  $B_a$  in the figure.
- The short intersecting curve represents the range of subjective brightness that the eye can perceive when adapted to this level. This range is rather restricted, having a level  $B_b$  at and below which all stimuli are perceived as indistinguishable blacks.
- The upper (dashed) portion of the curve is not actually restricted but, if extended too far, loses its meaning because much higher intensities would simply raise the adaptation level to a higher value than  $B_a$ .
- The long solid curve represents the range of intensities to which the visual system can adapt.
- In photopic vision alone, the range is about  $10^6$ .
- The transition from scotopic to photopic vision is gradual over the approximate range from 0.001 to 0.1 millilambert (-3 to -1 mL in the log scale), as illustrated by the double branches of the adaptation curve in this range.
- The visual system cannot operate over such a range simultaneously. It accomplishes this large variation by changes in its overall sensitivity. This phenomenon is known as brightness adaptation.
- The total range of intensity levels it can discriminate simultaneously is rather small compared with the total range.
- For any given set of conditions, the current sensitivity level of the visual system is called the brightness adaptation level.
- Two phenomena clearly demonstrate that perceived brightness is not simply a function of intensity
- [Mach Bands](#)
- [Simultaneous contrast](#)

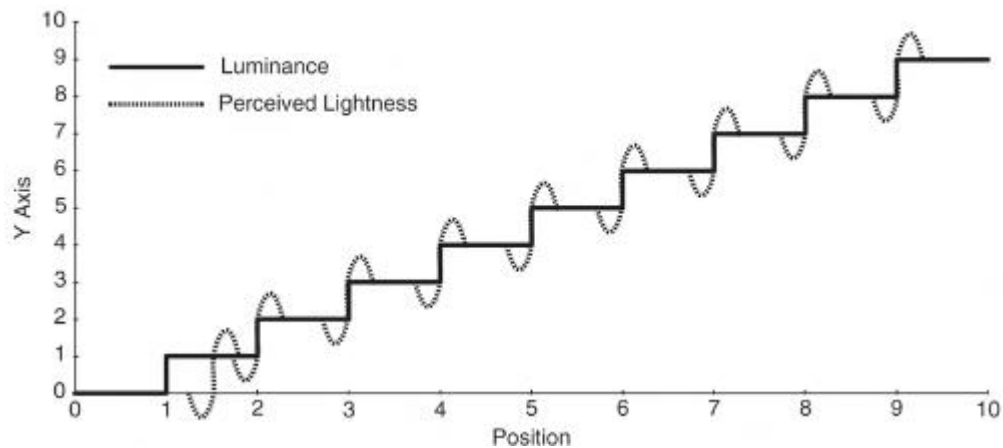
## Mach Bands

- The Mach Bands are the adjacently spaced rectangular stripes of constant intensities to demonstrate the phenomenon of simultaneous contrast.
- The Mach bands effect is due to the spatial high-boost filtering performed by the human visual system on the luminance channel of the image captured by the retina.
- **Along the boundary between adjacent shades of grey** in the Mach bands illusion, lateral inhibition makes the **darker area falsely appear even darker and the lighter area falsely appear even lighter**.



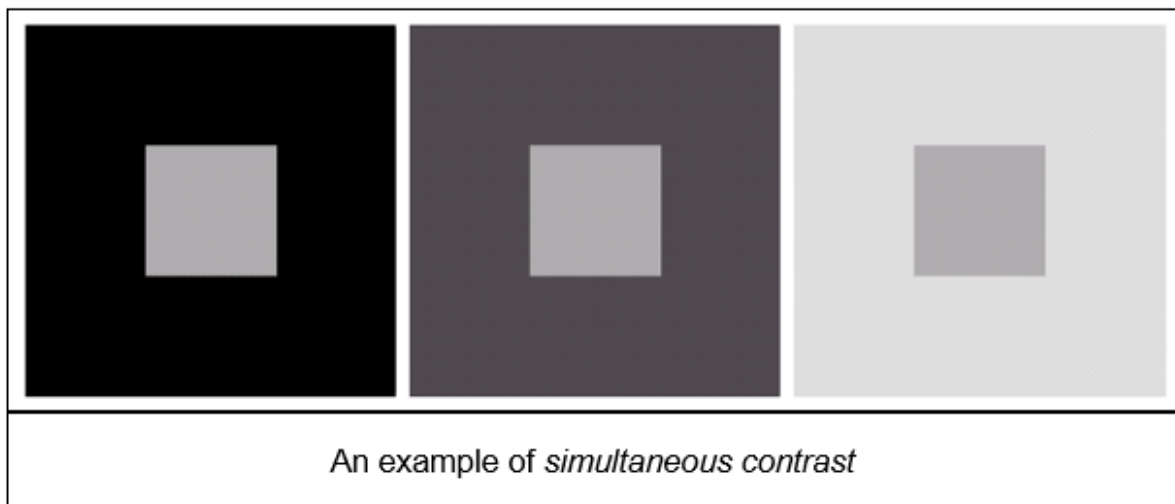
The visual system tends to undershoot or overshoot around the boundary of regions of different intensities

- In the case of visual information, such (spatial) differentiation causes gradual changes in the contrast between an object and its background to be enhanced, i.e., to become more visible.
- In human perception, this contrast enhancement produces what is known as Mach Bands: between two regions of different intensity a thin bright band appears at the lighter side and a thin dark band appears on the darker side.
- These bands are not physically present are "overshoot" and "undershoot" caused by our neural circuits in processing a step discontinuity in illumination.



## Simultaneous Contrast

- **The simultaneous contrast phenomenon: each small square is actually the same intensity but, because of the different intensities of the surrounds, the small squares do not appear equally bright.**
- The hue of a patch is also dependent on the wavelength composition of surrounding light.
- A white patch on a black background will appear to be yellowish if the surround is a blue light.



## Optical illusion

- An optical illusion (also called a visual illusion) is an illusion caused by the visual system and characterized by a visual percept that appears to differ from reality.
- The eye fills in non-existing information or wrongly perceives geometrical properties of objects

Our visual  
systems play lots  
of interesting  
tricks on us

